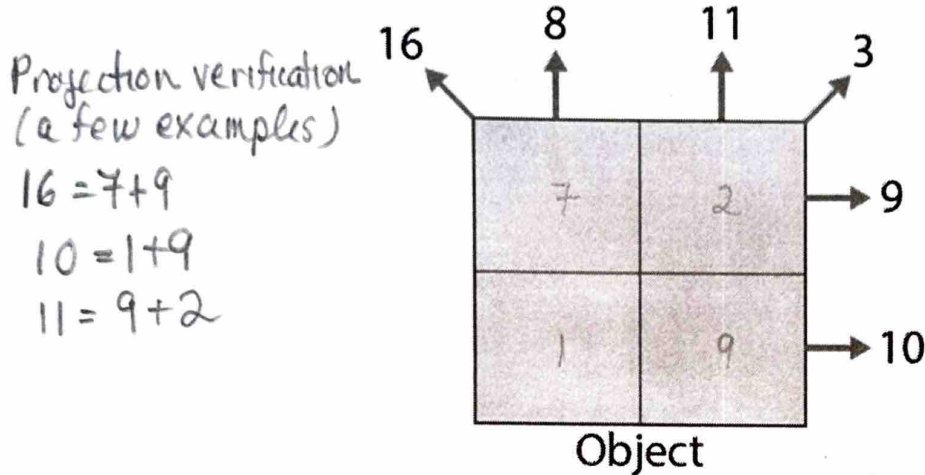


Name Key

Physics 390 In-Class Activity 12

Iterative Reconstruction

Complete the following tables using principles discussed in class to reconstruct the object based on the indicated activity projections:



Object is unknown but shown in light font so the projections and final reconstruction can be checked.

0	0
↑	↑
0	0
0	0

Guess

Use two vertical projections (8 & 11)

4	5.5
4	5.5

First Iteration

Corrections are $\frac{8-0}{2}$ and $\frac{11-0}{2}$
These are added to previous guess values (0)

4 + 3.25 = 7.25	5.5 - 3.25 = 2.25
4 - 3.25 = 0.75	5.5 + 3.25 = 8.75

Second Iteration

Use two angled projections (16 & 3)

7.25 - 0.25 = 7	2.25 - 0.25 = 2
0.75 + 0.25 = 1	8.75 + 0.25 = 9

Final Reconstruction

Use two horizontal projections

Corrections $\frac{9-9.5}{2}$ and $\frac{10-9.5}{2}$

Corrections $\frac{16-9.5}{2}$ and $\frac{3-9.5}{2}$